

ThermalCareBench: Auditable Multimodal Agent Evaluation for Thermal-Preference Support

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Abstract. Multimodal agents are increasingly asked to turn heterogeneous observations into recommendations, yet their evaluation rarely requires both evidence-grounded reasoning and explicit safety constraints. We introduce ThermalCareBench, an auditable benchmark for thermal-preference support from synchronized wearable, environmental, behavioural, and historical evidence. The benchmark uses a public dataset of 3,843 observations from 14 adults and exposes a partially observed, tool-mediated environment with frozen evaluator-only answers. Each trajectory is assessed for preference agreement, evidence sufficiency, action safety, tool budget, and scope violations. In a 280-instance frozen evaluation, systems with explicit evidence and policy constraints achieve substantially higher safety rates than unconstrained variants, while the strongest evaluated system attains 0.436 preference agreement and 0.975 safety. We further provide an independent wearable-data supplement and an API-free clean-room rerun. ThermalCareBench is deliberately non-clinical: it evaluates observed thermal-preference support, not diagnosis, heat-illness prediction, real-time outage detection, or population health effects.

Keywords: Multimodal agents · Benchmark evaluation · Thermal comfort · Tool use · AI safety

1 Introduction

Multimodal agents are increasingly expected to move beyond prediction and provide situated support. In personalized thermal-comfort settings, useful assistance may depend on environmental measurements, wearable physiological signals, sensor quality, and prior user preferences. Such a system therefore demands more than static predictions. It must gather evidence, handle missing modalities, and select a constrained, low-risk comfort action.

Existing personal comfort models typically formulate the problem as a fixed-input classification [1]. This protocol primarily measures predictive accuracy, but leaves several critical behaviours unchecked. Specifically, the standard setup does not verify evidence sufficiency, temporal constraints on user history, or uncertainty calibration against sensor quality. It also gives little visibility into unsafe or unsupported recommendations. As a result, label accuracy alone does not establish whether the system inspected sufficient evidence, calibrated uncertainty, or satisfied action constraints.

ThermalCareBench is introduced as an executable benchmark for comfort support tasks. It turns synchronized wearable and environmental observations into partially observed, tool-mediated tasks. Agents must retrieve evidence through typed tools and submit structured responses, while labels and scoring policies remain hidden in a frozen evaluator. This design evaluates the full evidence-to-action trajectory, separating preference agreement from evidence sufficiency, safety, tool use, and scope control. It reflects a bounded and human-centered target, as it evaluates observed thermal-preference support and constrained low-risk comfort actions.

This work makes three key contributions. First, we introduce a frozen multimodal evidence-to-action environment with explicit tool contracts and evaluator-only labels. Second, we separate predictive agreement from safety, evidence sufficiency, and scope discipline, making unsupported but plausible responses observable. Finally, we release reproducibility artifacts, including versioned input manifests, an independent wearable supplement, and a clean-room test run.

2 Related Work and Scope

2.1 Personal thermal comfort models

Personal thermal comfort research has shifted from population-level estimation toward occupant-specific prediction. Studies now use environmental measurements together with wearable signals, such as skin temperature and heart rate [1,2]. Most evaluations, however, still frame the problem as fixed-input classification against recorded preference votes. This setup compares predictors effectively. It gives less insight into whether a support system gathered enough evidence, respected temporal constraints, handled unreliable sensors, or selected a bounded action under uncertainty.

2.2 Multimodal agent benchmarks

Recent agent benchmarks evaluate tool use, planning, and multi-step interaction in interactive environments [3]. These benchmarks provide useful tests of general-purpose tool operation. Person-level physiological and environmental evidence rarely appears in synchronized form. Hidden preference labels, sensor-quality uncertainty, evidence sufficiency, and bounded comfort-support actions also receive limited coverage.

2.3 Benchmark positioning and scope

ThermalCareBench connects personal comfort modeling with agent evaluation. It addresses this gap by combining real-world physiological and preference observations, paired with a strict agent tool-use contract. Existing urban heat and infrastructure datasets often emphasize broad spatial mapping. ThermalCareBench uses annual visual context, such as satellite night-light imagery, only

as background evidence and penalizes agents that treat it as same-day exposure. The benchmark scope covers observed thermal-preference votes and bounded comfort-support actions, while strictly avoiding clinical diagnosis or verification of external public services.

3 Benchmark Design

3.1 Frozen instances and partial observability

ThermalCareBench is designed as a frozen benchmark for evaluating evidence-based thermal-preference support under partial observability. The benchmark is constructed from the synchronized wearable dataset [4], which contains 3,843 observations collected from 14 participants. Table 1 summarizes the fixed benchmark composition and clarifies the separation between the released Liu core data, the 280 core agent-evaluation instances, Task X, and the independent Choi supplement. Rather than exposing complete records, we release 280 public evaluation instances sampled from participant-level leave-one-subject-out (LOSO) splits.

Each public instance contains only the information available at prediction time. We formalize this separation as:

$$x_i^{\text{pub}} = (u_i, t_i, r_i, \mathcal{O}_i), \quad x_i^{\text{eval}} = (y_i, \mathcal{E}_i^*, \pi_i^*)$$

The first tuple is exposed during inference, while the second remains frozen inside the evaluator. Agents therefore cannot directly access complete participant records or future observations during inference.

For the core agent evaluation, this design extends thermal-preference prediction beyond direct classification into an evidence-acquisition and bounded-action problem. Task P remains a separate participant-level prediction setting, while the agent track evaluates whether partial evidence is gathered and used appropriately before a support action is produced. Every evaluated agent receives the same frozen public instances and hidden evaluator, ensuring reproducible comparison across different reasoning systems while preventing information leakage from hidden annotations or future observations.

Table 1. Data composition and scope separation (fixed build; Task X and Choi are not pooled with Liu core).

Component	Records	Subjects/scenarios	Role
Liu core data	3,843	14 participants	Released multimodal source
Core AgentBench	280	280 instances	Main agent evaluation
Task X	50	5 cities $\times$ 10 dates	Separate scope stress test
Choi supplement	2,090	18 participants	Independent wearable check

Figure 1 illustrates how ThermalCareBench turns partially observed multimodal evidence into a bounded support action under an evaluator-only label.

ThermalCareBench evidence-to-action pipeline

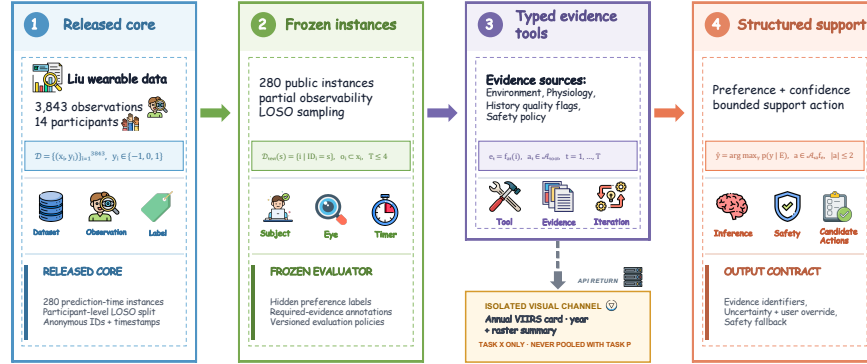


Fig. 1. ThermalCareBench evidence-to-action pipeline. The optional urban-context path is secondary and is not pooled with core preference results.

Figure 2 summarizes the released core cohort and the missingness patterns that motivate the benchmark’s evidence-acquisition design.

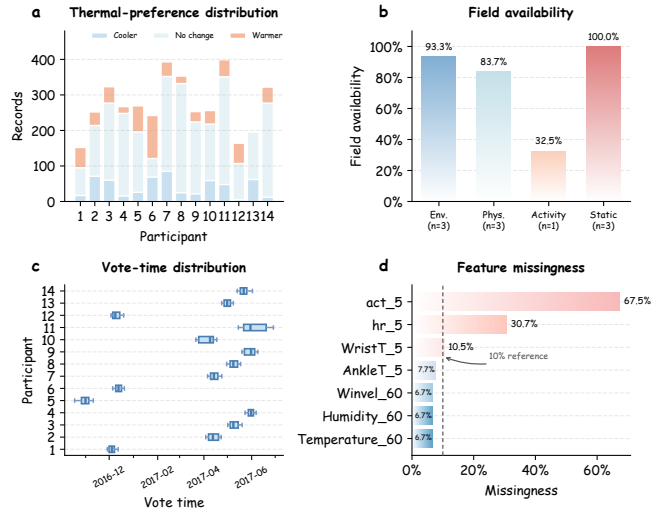


Fig. 2. Core data profile: preference-label composition by participant, field availability, vote-time coverage, and feature missingness.

3.2 Tools and output contract

Agents interact with ThermalCareBench exclusively through a fixed set of typed evidence tools [5,6]. These tools expose only the information that would realistically be available during thermal-preference support, including environmental observations, wearable physiology, temporally valid participant history, sensor-quality indicators, and a versioned non-clinical safety policy. Table 2 lists the complete typed tool contract, specifying each function name and the evidence it returns.

To ensure comparable reasoning behaviour across models, each evaluation instance allows at most four tool calls. This budget is enforced as a hard trajectory constraint:

$$\tau_i = (c_{i,1}, \dots, c_{i,m_i}), \quad m_i \leq B, \quad B = 4$$

The returned information is deterministic and frozen for every instance, preventing external knowledge or online data from influencing benchmark results.

The final response follows the structured output contract:

$$\hat{z}_i = (\hat{y}_i, p_i, a_i, \mathcal{I}_i, q_i, o_i, f_i)$$

The ordered fields represent the predicted thermal preference, confidence score, bounded support action, evidence identifiers, uncertainty explanation, optional user override, and safety fallback. Because every output field follows a predefined schema, evidence sufficiency, policy compliance, and safety violations can be evaluated automatically without requiring access to the model’s internal reasoning process.

Table 2. Typed tool contract available to the agent (fixed interfaces).

Tool type	Function	Returned evidence
Environment	<code>get_environment_summary</code>	temperature, humidity, context window
Physiology	<code>get_physiology_summary</code>	heart-rate and wearable summaries
History	<code>get_history(k)</code>	strictly-prior preference records
Quality	<code>get_quality_flags</code>	missingness and reliability flags
Safety policy	<code>get_safety_policy</code>	admissible actions and fallback rule
Urban context	<code>get_urban_context</code>	Task X annual imagery descriptor

3.3 Metrics and separation of tasks

ThermalCareBench evaluates agents from complementary perspectives rather than using a single composite score. Core evaluation focuses on preference agreement, evidence sufficiency, safe-action rate, average tool usage, and policy violation rate. For a frozen core set, the primary rates are computed as:

$$\begin{aligned} \text{Agree} &= \frac{1}{N} \sum_{i=1}^N \mathbf{1} [\hat{y}_i = y_i], \\ \text{Evidence} &= \frac{1}{N} \sum_{i=1}^N \mathbf{1} [\mathcal{E}_i^* \subseteq \mathcal{I}_i], \\ \text{Safe} &= \frac{1}{N} \sum_{i=1}^N \mathbf{1} [a_i \in \mathcal{A}_{\pi_i^*}]. \end{aligned}$$

Average tool use is reported separately from these rates.

Task X is evaluated independently as an urban-context stress test containing 50 frozen scenarios. Unlike the core benchmark, Task X assesses temporal consistency, external-tool discipline, and semantic scope control when interacting with weather, routing, and contextual information. Since these signals are not participant-specific, Task X scores are never aggregated with the core thermal-preference metrics. Accordingly, the reported evaluation sets are disjoint by construction:

$$\mathcal{D}_{\text{core}} \cap \mathcal{D}_X = \emptyset, \quad \mathcal{D}_{\text{core}} \cap \mathcal{D}_{\text{Choi}} = \emptyset$$

Separating the two evaluation tracks prevents contextual tools from receiving predictive credit for participant-level preference prediction while enabling complementary analysis of reasoning quality and external-tool robustness.

ThermalCareBench contains two complementary evaluation components. Task P evaluates participant-level thermal-preference prediction from synchronized multimodal observations, for which we introduce QAPCF as a quality-aware predictive method. The core agent evaluation assesses whether an agent can transform partially observed evidence into a bounded, policy-compliant support response. These components address different levels of the same evidence-to-action problem and are reported separately.

4 Experimental Protocol

4.1 Evaluation systems

We evaluate five controlled agent configurations (A1–A5) on the same 280 public benchmark instances. A4 is the complete proposed agent system, combining tool-mediated evidence acquisition with the structured policy-constrained support contract. A1–A3 serve as unconstrained component-ablation baselines that isolate the effects of evidence access and policy constraints. A5 is an oracle diagnostic baseline with predefined privileged evidence access; it is included to characterize the residual gap under non-deployable conditions and is not the proposed system. The configurations differ only in the evidence sources and policy constraints available to the agent, while sharing an identical evaluator, prompt template, and scoring protocol. This controlled setting isolates the effect of evidence acquisition and policy enforcement without introducing confounding differences in the evaluation procedure.

All systems are executed using the same frozen API model snapshot. Hidden answer files remain inaccessible during inference and are used exclusively by the evaluator after generation, ensuring that every configuration is evaluated under identical benchmark conditions.

4.2 Evaluation Procedure

Each system processes the same frozen benchmark instances independently. During evaluation, the agent may interact with the released evidence tools under the predefined tool budget and output the structured support response specified in

Section 3. For A1–A4, evidence is accessed only through the released tool interfaces under the predefined tool budget. A5 follows its separately specified oracle diagnostic condition and is scored with the same frozen evaluator, but it is not interpreted as a tool-mediated deployable trajectory.

After inference is completed, the evaluator compares the submitted outputs with the hidden answer file and computes all benchmark metrics automatically, including preference agreement, evidence sufficiency, safe-action rate, tool usage, and policy violation rate. Because the evaluator is identical for all systems, the reported differences reflect only the behaviour of the evaluated agents rather than differences in the evaluation process.

4.3 Predictive Baseline

Besides the agent benchmark, we retain a compact predictive baseline for Task P to provide reference preference-prediction performance under the same participant-level evaluation protocol. The baseline uses the proposed QAPCF model and is evaluated independently from the agent benchmark using leave-one-subject-out (LOSO) validation. Table 3 reports the balanced accuracy, Macro-F1, and accuracy of QAPCF alongside the predictive baselines under the LOSO protocol.

Task P and the core agent track answer different evaluation questions. Task P evaluates the predictive and uncertainty-aware properties of QAPCF against predictive baselines and ablations, whereas the agent track evaluates evidence acquisition, structured support generation, and policy compliance. Their metrics are therefore reported separately and are not used to rank QAPCF against agent configurations.

Table 3. Task P results under the frozen evaluation protocol. Balanced accuracy and Macro-F1 are computed over the three thermal preference classes.

Model	Balanced accuracy	Macro-F1	Accuracy
B5	0.422	0.398	0.656
B2	0.452	0.386	0.534
QAPCF	0.510	0.368	0.436
B4	0.373	0.329	0.670
B3	0.332	0.267	0.655
B1	0.333	0.262	0.666

Figure 3 shows the QAPCF baseline pipeline, which mirrors the benchmark’s emphasis on separating uncertainty estimation from the final bounded action selection.

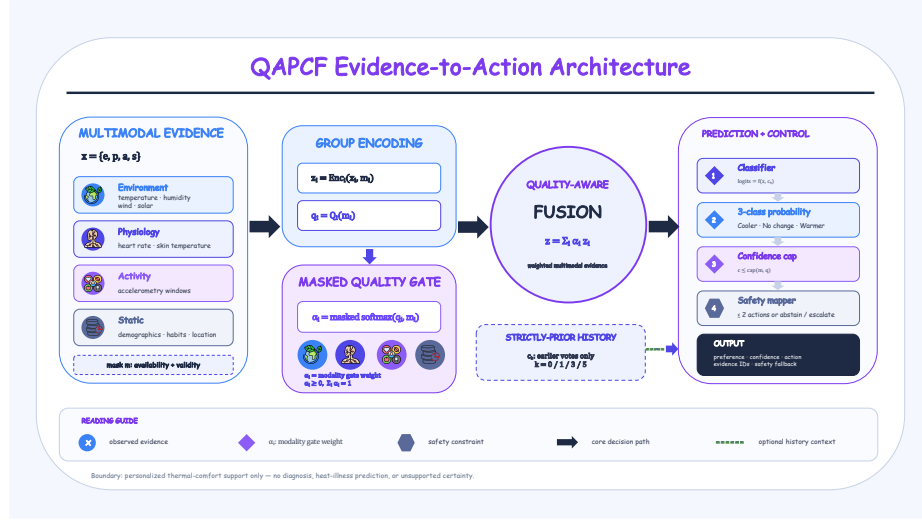


Fig. 3. QAPCF baseline protocol: quality-aware fusion produces frozen out-of-fold probabilities and uncertainty, which are then consumed by the constrained action layer.

5 Core Results

5.1 Overall Benchmark Performance

Figure 4 summarizes the overall benchmark results. The unconstrained configurations A1 and A2 achieve non-zero preference agreement, yet neither produces evidence-sufficient or safe support actions. This contrast illustrates the benchmark’s central unsupported-but-plausible failure mode: agreement with an observed preference vote alone does not establish that a response was evidence-grounded or policy-compliant.

Among deployable tool-mediated configurations, A4 is the complete proposed system. It achieves 0.411 preference agreement, 1.000 evidence sufficiency, a 0.961 safe-action rate, and a 0.039 policy-violation rate. Relative to A1–A3, these results show that combining structured evidence acquisition with policy constraints substantially improves the reliability of the resulting support behaviour.

A5 is an oracle diagnostic baseline rather than a deployable agent configuration. Its 0.436 agreement and 0.975 safe-action rate characterize the remaining gap under privileged evidence conditions; these values should not be interpreted as the performance of the proposed system.

Table 4 reports the full agent-level results across all five configurations, including preference agreement, evidence sufficiency, safe-action rate, mean tool calls, and policy violation rate.

These observations indicate that evidence availability and policy enforcement primarily alter the behavioural characteristics of an agent rather than simply improving predictive accuracy. The benchmark therefore exposes complementary

aspects of agent quality that cannot be represented by preference agreement alone.

Table 4. Agent-level results. Preference agreement is measured on all instances; evidence sufficiency and safety are measured against the structured support contract.

System	Agreement	Evidence	Safe	Tools	Violations
A1	0.361	0.000	0.000	0	1.000
A2	0.354	0.000	0.000	0	1.000
A3	0.325	1.000	0.839	4	0.161
A4	0.411	1.000	0.961	4	0.039
A5	0.436	1.000	0.975	0	0.025

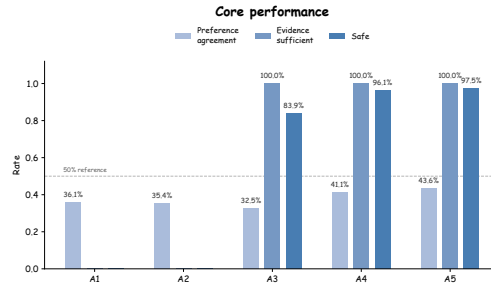


Fig. 4. Core results across the 280 frozen instances. Task X and the external wearable supplement are not pooled with these metrics.

5.2 Evidence acquisition changes the failure mode

Figure 5 further explains why the system configurations differ. A3 and A4 consistently consume the full four-call evidence budget to satisfy the required evidence contract. This pattern reflects the cost of obtaining auditable support from partial observations in the deployable tool-mediated setting.

A5 should not be interpreted as evidence that interactive evidence acquisition is unnecessary. It is an oracle diagnostic baseline operating under a predefined privileged-evidence condition rather than a comparable tool-use policy. Its role is to indicate the residual performance gap after evidence-access limitations are relaxed.

This behaviour should not be interpreted as evidence that information acquisition is unnecessary. Instead, it demonstrates that preference agreement alone cannot distinguish between evidence-supported and unsupported decisions. Reporting evidence sufficiency and policy compliance together with agreement is therefore essential for evaluating trustworthy thermal-preference support.

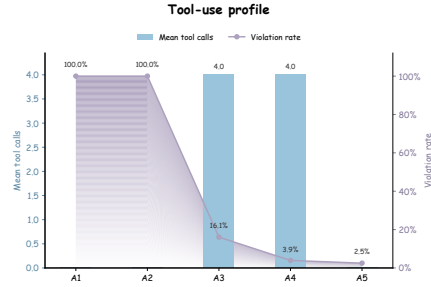


Fig. 5. Tool-use and violation measures for the core systems. The evaluator scores policy compliance separately from agreement with the recorded preference vote.

5.3 Predictive baseline for Task P

Task P evaluates QAPCF, our proposed quality-aware method for participant-level thermal-preference prediction, under the LOSO protocol. QAPCF is compared with B1–B5 baselines and ablations using balanced accuracy, Macro-F1, coverage, calibration, and selective-risk analyses.

QAPCF is designed not only to produce a preference prediction, but also to expose the quality and uncertainty of the predictive signal before it is consumed by a bounded action layer. Accordingly, its evaluation emphasizes the trade-offs among predictive performance, coverage calibration, history sensitivity, and selective risk rather than claiming uniform dominance on every scalar metric as shown in Figure 6.

Under LOSO evaluation, QAPCF achieves 0.510 balanced accuracy, 0.368 Macro-F1, and 0.436 accuracy. While individual baselines attain high values on particular scalar metrics, QAPCF provides the quality-aware fusion and uncertainty-aware outputs required by the prediction component of Thermal-CareBench.

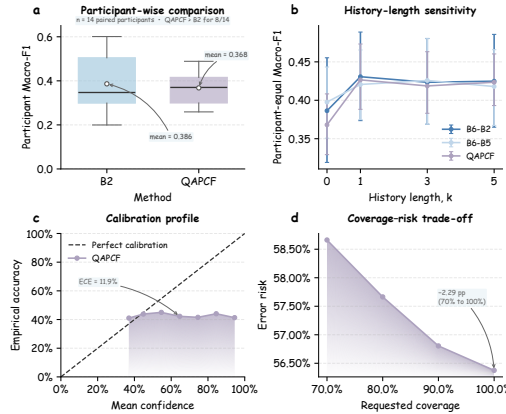


Fig. 6. Task P results: participant-level macro-F1, strictly-prior history curve, reliability diagram, and selective-risk curve. These results do not claim that QAPCF uniformly outperforms the baselines.

6 Independent Wearable Supplement

To evaluate whether the benchmark design remains portable beyond the core cohort without altering the primary benchmark, we include an independent public wearable dataset from Choi et al. [7]. This supplementary cohort is evaluated separately and is never aggregated with the core ThermalCareBench metrics.

The cohort contains 2,090 synchronized wearable observations collected from 18 participants, including electrodermal activity, skin temperature, heart rate, and three-state thermal-preference labels. Following the same participant-level evaluation protocol, we train a standardized multinomial logistic-regression baseline under leave-one-subject-out (LOSO) validation.

As summarized in Table 5, the baseline achieves a balanced accuracy of 0.422 and a Macro-F1 of 0.427, exceeding the one-third chance reference. Figure 7 further reports the label distribution and baseline performance for this supplementary cohort.

These results should not be interpreted as an extension of the core benchmark ranking or as evidence of cross-dataset transfer. Instead, they provide an independent cohort-specific reference demonstrating that the benchmark protocol can be instantiated on an external wearable dataset while remaining analytically separate from the primary ThermalCareBench evaluation.

Table 5. Independent wearable supplement. This cohort is reported separately and is not combined with the Liu core benchmark metrics.

Cohort	Held-out records	Subjects	Balanced accuracy	Macro-F1
Choi et al. wearable cohort	2,090	18	0.422	0.427

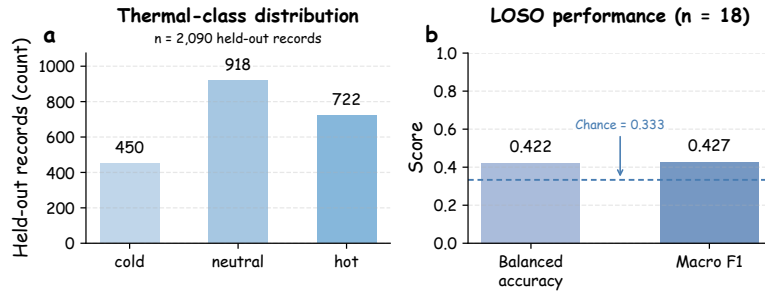


Fig. 7. Independent wearable supplement. Counts and LOSO baseline scores are shown for the Choi cohort only; they are not combined with core benchmark metrics.

7 Reproducibility, Limitations, and Ethics

Because ThermalCareBench evaluates complete evidence-to-action trajectories, reproducibility is part of the benchmark design rather than a post hoc artifact. The reproducibility release includes frozen inputs, code, source manifests, score reports, and an API-free clean-room run. The clean-room run validates the

pipeline without live model access in a fresh virtual environment with pinned dependencies. The external wearable data were obtained independently from the published source, and their provenance is recorded in the release manifest.

Several limitations follow from the same bounded design. The core dataset is small and observational, so benchmark outcomes cannot support claims about clinical efficacy or population generalization. The reported configuration also precludes cross-model comparison, since the completed model run uses one fixed API model snapshot. Task X serves only as a context-discipline test; it does not validate facility availability, real-time electricity status, or health outcomes.

All analyses use public research data, and the release contains only anonymized instance tokens. The benchmark action space is limited to low-risk comfort-support options, with explicit paths for user override and human escalation. ThermalCareBench serves as an evaluation benchmark. It does not provide medical advice, clinical decision support, emergency guidance, or verification of external public services.

8 Conclusion

ThermalCareBench provides an executable, auditable framework for evaluating how multimodal agents use evidence and constraints in thermal-preference support. Rather than collapsing performance into a single accuracy score, the benchmark reports preference agreement, evidence sufficiency, safety, tool use, and scope compliance separately. This decomposition exposes failure modes that accuracy-only evaluations can obscure, particularly responses that appear plausible without sufficient supporting evidence.

For assistive systems, trustworthiness depends not only on matching an observed label but also on acquiring relevant evidence, responding cautiously under uncertainty, and linking each action to its supporting observations. ThermalCareBench therefore evaluates both the prediction and the process that produces a bounded support action.

The same audit-oriented design may extend to personal-assistance domains characterized by partial information, latent preferences, and low-risk interventions. The frozen run, independent wearable supplement, and clean-room reproducibility artifacts provide a reproducible basis for future comparisons of evidence-aware and user-autonomy-respecting multimodal agents.

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